

<sol_start id=1>

1. (1) 3

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<sol_start id=2>

2. (3) Infinitely many solutions.

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<sol_start id=3>

3. $\sqrt{(-6-0)^2(8-0)^2}$

$$= \sqrt{36 + 64}$$

$$\sqrt{100}$$

$$= 10$$

(1) 10 units

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<sol_start id=4>

4. (2) proportional

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<sol_start id=5>

5. (1) $\frac{1}{2}$

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<sol_start id=6>

6. (1) $x^3 y^3$

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7. Let the roots be α & β , $\alpha + \beta = \frac{1}{2} \times \beta$

$$\frac{-(-2)}{4} = \frac{1}{2} \times \frac{(K-4)}{4}$$

$$2 = \frac{1}{2} \times \frac{K-4}{4}$$

$$4 = \frac{K-4}{4}$$

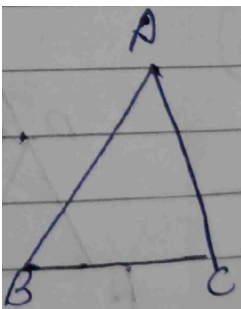
$$16 = K - 4$$

$$\boxed{K = 20}$$

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8. Let ABC be the equilateral triangle.



$$A = (0,0)$$

$$B = (3, \sqrt{3})$$

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(3-0)^2 + (\sqrt{3}-0)^2}$$

$$= \sqrt{9+3}$$

$$= \sqrt{12}$$

AB = BC So.

$$BC \sqrt{(3-3)^2 + (x-\sqrt{3})^2} = \sqrt{12}$$

$$(x-\sqrt{3})^2 = 12$$

$$x^2 - 2\sqrt{3}x + 3 = 12$$

$$x^2 - 2\sqrt{3}x + 9 = 0$$

$$x^2 + \sqrt{3}x - 3\sqrt{3}x - 9$$

$$x(x - \sqrt{3}) - 3\sqrt{3}(x + \sqrt{3})$$

$$(x + \sqrt{3})(x - 3\sqrt{3})$$

$$BC = (3\sqrt{3} - \sqrt{3})$$

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$$9. \quad \text{no. solution} = \frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$

$$\text{So, } \frac{2p}{2} = \frac{3}{6}$$

$$= \frac{2p}{2} = \frac{1}{2}$$

$$p = \frac{1}{2}$$

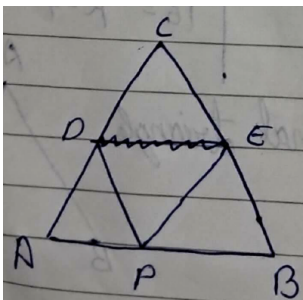
$\therefore$ P can be any value that then $\frac{1}{2}$

So it be only one solution.

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10.



$PD \parallel BC$

$$\text{Given} \rightarrow \frac{AD}{CD} = \frac{AP}{BP} \quad [\text{BPT} - (1)]$$

$$\text{Also given} \rightarrow \frac{AD}{CD} = \frac{CE}{BE} \quad -(2)$$

From (1) & (2)

$$\frac{AD}{CD} = \frac{AP}{BP} = \frac{CE}{BE}$$

$$\text{So} \quad \frac{BE}{CD} = \frac{BP}{AP}$$

$\therefore PE \parallel AC$ [Converse of BPT]

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$$11. \quad \tan \theta = \frac{1}{2}$$

$$\text{To prove} \quad \frac{1}{2} \sin \theta \cdot \cos \theta = \frac{1}{5}$$

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{2}$$

$$\sin \theta = \frac{1}{2} \cos^2 \theta$$

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$$12. \quad \frac{257}{5000} = \frac{257}{3^2 \times 5^4}$$

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13. Let us assume $\sqrt{3}$ is a rational number.

$$\text{ie } \sqrt{3} = \frac{a}{b} \quad [\text{where } a \text{ \& } b \text{ are co - primes \& } b \neq 0]$$

$$\sqrt{3} \cdot b = a \text{ [squaring both sides]}$$

$$3b^2 = a^2$$

3 divides $a^2 \rightarrow$ 3 divides a also

$$\text{i.e. } a = 3c$$

$$3b^2 = (3c)^2$$

$$\cancel{3}b^2 = \cancel{3}c^2$$

$$b^2 = 3c^2$$

$\therefore b^2$ has more than one factor.

i.e. our assumption is wrong.

$\sqrt{3}$ is an irrational number.

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$$15. \quad 3x - (a + 1)y = 2b - 1$$

$$5x + (2 - 2a)y = 3b$$

$$\text{Infinitely many solutions} \rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{3}{5} = \frac{-(a+1)}{1-2a} = \frac{2b-1}{3b}$$

$$\frac{3-6a-5a+5}{5-10a} = \frac{2b-1}{3b}$$

$$3 - 6a = -5a + 5$$

$$a + 2 = 0$$

$$a = -2$$

$$\frac{3 \times 6a}{-5a + 5}$$

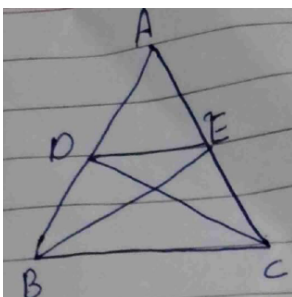
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16.



Given $\triangle ABE \cong \triangle ACD$

i.e. $AB = AC$ [c.p.c.t]

also $AD = AE$ -(1)

In $\triangle ADE$ & $\triangle ABC$

$CA = CA$ [common]

$AD = DE$ [Fron (1)]

$\therefore \triangle ABC$ & $\triangle ADE$ [SAS similarity]

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$$17. \quad \frac{\left(\frac{\sin \theta}{\cos \theta}\right)^2}{\sec^2 \theta} + \frac{1 - \cot^2 \theta}{1}$$

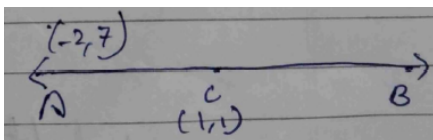
$$\frac{\left(\frac{\sin \theta}{\cos \theta}\right)^2}{\sec^2 \theta} + \frac{1 - \cot^2 \theta}{\sin \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

$$1 + \tan^2 \theta$$

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<sol_start id=18>

$$18. \quad \text{section formula} \rightarrow \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$



$$\left(\frac{3x_2 + 2(-2)}{3+2}, \frac{3y_2 + 2(7)}{3+2} \right)$$

$$\frac{3x_2 - 4}{5} = 1$$

$$3x_2 - 4 = 5$$

$$3x_2 = a$$

$$x_2 = \frac{a}{3}$$

$$x_2 = 3$$

$$\frac{3y_2 + 14}{5} = 1$$

$$3y_2 + 14 = 5$$

$$3y_2 = 5 - 14$$

$$3y_2 = -9$$

$$y_2 = -3$$

$$\text{therefore } B = (3, -3)$$

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19. Let the numerator & denominator be x & y respectively

$$\frac{2x}{y+1} = 1$$

$$2x = y + 1$$

$$2x - y - 1 = 0 \quad -(1)$$

Case (2)

$$\frac{x+4}{y(2)} = 1$$

$$= x - 4 - 2y$$

$$= x - 2y + 4 = 0 \quad -(2)$$

$$(1)-(2)$$

$$2x - y - = 0 - 1$$

$$-2x - 4y + = 0 - 4$$

$$\hline 3y = -5$$

$$y = \frac{-5}{3}$$

$$\text{substitute } y = \frac{-5}{3} \text{ in eq. (2)}$$

$$x - 2 \left(\frac{-5}{3} \right) = 4$$

$$x - \frac{10}{3} = 4$$

$$x = 4 + \frac{10}{3}$$

$$x = \frac{12+10}{3}$$

$$= \frac{22}{3}$$

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20. Basic proportionality theorem →

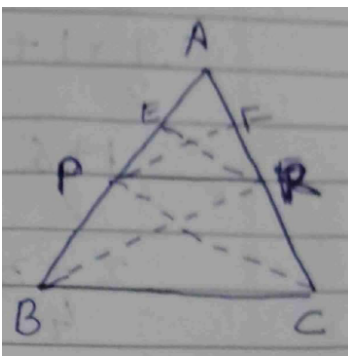
If a line draw parallel to one under of the triangle

Given .

$PR \parallel BC$

To prove

$$\frac{AP}{BP} = \frac{AR}{CR}$$



Construction → Draw PF & ER

Draw BR & PC

$$\text{Proof Ar. } \triangle APR = \frac{1}{2} \times AP \times ER \quad -(1)$$

$$\text{Ar. } \triangle BPE = \frac{1}{2} \times BP \times ER \quad -(2)$$

$$\frac{(1)}{(2)} \rightarrow \frac{\frac{1}{2} \times AP \times \cancel{ER}}{\frac{1}{2} \times BP \times \cancel{ER}}$$

$$= \frac{AP}{BP} \quad -(3)$$

$$\text{Ar. } \triangle ARP = \frac{1}{2} \times AR \times PF \quad -(4)$$

$$\text{AR. } \triangle PRC = \frac{1}{2} \times CR \times PE \quad -(5)$$

$$\frac{(4)}{(5)} \rightarrow \frac{\frac{1}{2} \times AR \times \cancel{PF}}{\frac{1}{2} \times CR \times \cancel{PF}} = \frac{AR}{CR} \quad -(6)$$

From (3) & (6)

$$\frac{AP}{BP} = \frac{AR}{CR}$$

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$$21. \quad \frac{\sin 30^\circ + 2\cos^2 45^\circ + \tan^2 60^\circ}{\frac{1}{2}\cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$$

$$\frac{\frac{1}{2} + 2 \times \frac{1}{2} + 3}{\frac{1}{2} \times 1 + \frac{3}{4} + 1^2}$$

$$= \frac{\frac{1+6}{2}}{\frac{2+3}{4} + 1}$$

$$\frac{\frac{7}{2}}{\frac{5}{4}+1}$$

$$\frac{\frac{7}{2}}{\frac{5+4}{4}}$$

$$\frac{\frac{7}{2}}{\frac{9}{4}}$$

$$\frac{7}{2} \times \frac{4}{9}$$

$$\frac{14}{9}$$

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22. AB = CD [opposite sides]

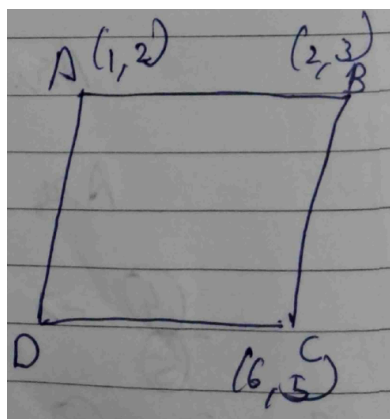
Distance formula

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AB = \sqrt{(2-1)^2 + (3-2)^2}$$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$



$$CD = \sqrt{(x_2 - 6)^2 + (y_2 - 5)^2} = \sqrt{2}$$

$$(x_2 - 6)^2 + (y_2 - 5)^2 = 2$$

$$x^2 - 12x + 36 + 25 + y^2 - 10y = 2$$

$$x^2 + y^2 + 61 - 12x - 10y = 2$$

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<sol_start id=23>

23. LCM of 330, 198 & 220

$$330 = 10 \times 3 \times 11 = 2 \times 3 \times 5 \times 11$$

$$198 = 2 \times 3 \times 3 \times 11 = 2 \times 3^2 \times 11$$

$$220 = 10 \times 2 \times 11 = 2^2 \times 5 \times 11$$

$$\begin{aligned} \text{LCM}(330, 198, 220) &= 5^1 \times 2^2 \times 11 \times 3^2 \\ &= 5 \times 4 \times 11 \times 9 \\ &= 1980 \text{ min.} \end{aligned}$$

After 33 hours they will be together.

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24. Let the father's age be x & son's age by y .

$$2y + x = 70 \quad \text{----(1)}$$

$$2x + y = 95 \quad \text{-----(2)}$$

$$x + 2y = 70$$

$$x + y = 95$$

adding by (1) by (2)

$$\cancel{2x} + 4y = 140$$

$$\underline{-\cancel{2x} + y = 95}$$

$$3y = 45$$

$$y = \frac{45}{3}$$

$$y = 15$$

superstition $y = 15$ in by (2)

$$2x + 15 = 95$$

$$2x = 80$$

$$X = 40$$

Father's age $\rightarrow 40$

Son's age $\rightarrow 15$

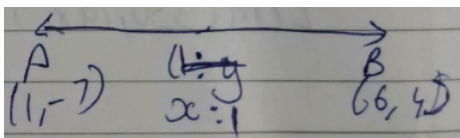
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25. Section formula



$$\rightarrow \left(\frac{m_2 x_1 + m_1 x_2}{m_1 + m_2}, \frac{m_2 y_1 + m_1 y_2}{m_1 + m_2} \right)$$

$$\rightarrow 1(1) + x$$

<sol_end>

<sol_start id=26>

26. $\operatorname{cosec} \theta - \sin \theta = m$, $\sec \theta - \cos \theta = n$

$$\left((\operatorname{cosec} \theta - \sin \theta)^2 (\sec \theta - \cos \theta) \right)^{\frac{2}{3}} +$$

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27. (i) 3 : 2

(ii) 15 miles

(iii) $\frac{10}{7}$ miles

(iv) 35 miles

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<sol_start id=28>

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28. (i) Let (x_1, y_1) be $(2, 5)$

(x_2, y_2) be $(8, 3)$

ratio $\rightarrow 1 : 2$

section formula $\rightarrow \left(\frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}, \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} \right)$

$$= \left(\frac{1(2) + 2(8)}{1 + 2}, \frac{1(5) + 2(3)}{1 + 2} \right)$$

$$= \left(\frac{10}{3}, \frac{13}{3} \right)$$

$$= \left(4, \frac{13}{3} \right)$$

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(ii) Distance formula $\rightarrow \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Let (x_1, y_1) be $(2, 5)$

(x_2, y_2) be $(8, 3)$

$$\sqrt{(8 - 2)^2 + (3 - 5)^2}$$

$$\sqrt{36 + 4}$$

$$\sqrt{40} \text{ units}$$

<sub_id end>

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(iii) 2 units

<sub_id end>

<sub_id = 4>

(iv) Multipoint formula $\rightarrow \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

$$= \left(\frac{2+8}{2}, \frac{5+3}{2} \right)$$

$$= \left(\frac{10}{2}, \frac{8}{2} \right)$$

$$= (5, 4)$$

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$$(II) \quad \tan \theta = \frac{1}{2}$$

$$= \frac{\sin \theta}{\cos \theta} = \frac{1}{2}$$

$$= \sin \theta = \frac{1}{2}, \cos \theta$$

$$= \sin^2 \theta = \frac{1}{2}, \cos \theta, \sin \theta$$

$$\tan \theta = \frac{\theta_{p-p}}{\text{Adj.}}$$

$$1^2 + 2^2 = H^2$$

$$1 + 4 = H^2$$

$$5 = H^2$$

$$H = \sqrt{5}$$

$$\left(\frac{1}{\sqrt{5}} \right)^2 = \frac{1}{2} \sin \theta \cos \theta$$

$$\frac{1}{2} \times \sin \theta \cos \theta = \frac{1}{5}$$

$$\sin \theta = \frac{\text{Ops}}{\text{hypotenuse}}$$

$$= \frac{1}{\sqrt{5}}$$

Hence formula

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